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Development of a rotary magnetic refrigerator

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ARTICLE INFO

Article history:

Received 19 August 2009

Received in revised form

23 October 2009

Accepted 2 November 2009

Available online 10 November 2009

Keywords:

Magnetic refrigerator

Design

Construction

Magnet

Gadolinium

ABSTRACT

A team of researchers have developed and built a prototype of a rotary magnetic refrigerator (MR). The principle of the operation of the presented magnetic refrigerator is based on the rotary movement of active magnetic regenerators (AMRs) with a magnetocaloric material and on the magnetic field generated by permanent magnets Nd–Fe–B. The first part of the paper presents the development and analysis of the structure for generating the magnetic field and the basic operational principle of the magnetic refrigerator concerned, and the second part gives a description of the development process for certain key elements of the complete magnetic refrigeration system.

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Développement d'un réfrigérateur magnétique rotatif

Mots clés : Réfrigérateur magnétique ; Conception ; Construction ; Aimant ; Gadolinium

1. Introduction

Although the development of magnetic refrigeration started with the discovery of the magnetocaloric effect by Warburg (1881), two other events brought major advances to the technology of magnetic refrigeration at room temperature. The first of these events was the design of the first magnetic room temperature refrigerator developed by Brown (1976). The second milestone was year 1997, when Pecharsky and Gschneidner (1997) discovered the giant magnetocaloric effect in the $Gd_5Si_2Ge_2$ alloy, which is approximately 50% higher than the magnetocaloric effect of Gd metal. The discovery of the giant magnetocaloric effect spurred a broad international interest in

the field of magnetic refrigeration. Since this groundbreaking discovery, the development of prototype magnetic refrigerators has spread to numerous universities and institutes across the globe. More than 25 prototype magnetic refrigerators capable of operating with a varying degree of efficiency at room temperature have been built and tested to date. The prototypes show a cooling power of up to 100 W and a temperature span of up to 30 °C, depending on the magnetic flux density and the amount of magnetocaloric material. An intriguing group of devices, which attest to the fact that magnetic refrigeration could develop into a viable alternative to the vapour-compression refrigeration technology. The most advanced of all the prototypes are the three devices made by Zimm et al. (1998, 2006, 2007) which seem

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doi:10.1016/j.ijrefrig.2009.11.003

to be setting the trend of magnetic refrigeration development. A full review of room temperature magnetic refrigerators was undertaken by Gschneidner and Pecharsky (2008).

The magnetic refrigerator developed by the Laboratory of Refrigeration and the Laboratory for Modeling Machine Elements and Structures (both Faculty of Mechanical Engineering, University of Ljubljana) is based on the rotary movement of active magnetic regenerators (AMRs) placed in a rotating drum and the magnetic field generated by permanent magnets Nd-Fe-B. Through the use of connecting elements made of soft ferromagnetic material, four permanent magnets with focus elements inside and outside the rotating drum create a relatively strong and homogenous magnetic field in the air gaps. On the circumference of the rotating drum, four air gaps interchange with four slightly wider demagnetization areas with a low magnetic field.

2. Structure for generating magnetic field

The structure for magnetic field generation in the developed magnetic refrigeration device is designed on the basis of two magnetic circuits. Its geometry was chosen in close consideration of the financial aspect and future trends in magnetic refrigeration which seem to focus on developing rotating magnetic refrigerators with several regions of large magnetic fields (see Gschneidner and Pecharsky, 2008). Such solutions ensure a higher operating frequency of the magnetic refrigerator as the magnetocaloric material is exposed to several processes of magnetization and demagnetization in one revolution of the rotating drum. However, a higher operating frequency can be established only when excellent heat transfer and fluid dynamics characteristics exist in AMRs.

In the first step of the process, the geometry of the structure was optimized and a numerical simulation of the magnetic field generated by the magnet structure was carried out. In order to verify the obtained numerical results, the magnetic field was then measured to identify the exact values of magnetic flux density.

2.1. Numerical simulation

As there are several commercial programs that can solve very complex problems without the need to develop specific algorithms, the finite-element method is currently the most accessible and widely used for conducting a numerical simulation of the magnetic field. On account of its simple use, the FEMM (Finite-Element Method Magnetics, version 4.0) program, able to solve two-dimensional and low-frequency electromagnetic problems (see Meeker, 2006), was used to simulate the magnetic field which is generated by the structure containing permanent magnets and a soft ferromagnetic material. First, we needed to optimize the geometry to ensure that the minimum amount of material is used and that the magnetic field is strongest in areas where the magnetocaloric material is to be magnetized and weakest in areas where the magnetocaloric material is to be demagnetized. Additionally, the height of the air gaps was optimized in order for the gaps to meet the purpose of the research: on the one hand, they have to be as small as possible to generate higher magnetic flux density; on

the other hand, the air gaps have to contain the maximum amount of magnetocaloric material in order to increase the cooling power of the refrigerator. The optimization was performed by plotting charts of average magnetic flux density in the air gap as a function of different dimensions of the basic parts of the structure (the width and the height of the magnets, the thickness of the external ring and the height of the air gaps), which was worked out by Tušek (2007). The scheme of the structure with optimized dimensions is shown in Fig. 1 in a two-dimensional form. The length of the structure in axial direction is 170 mm (external ring) and 90 mm (magnets and inner yoke), while in the area of air gaps it is reduced to 55 mm (isometric dimensions in Fig. 1). The magnet structure is designed on the basis of four neodymium-iron-boron permanent magnets (Nd-Fe-B with 40 MGOe maximum energy product). These magnets are currently among the strongest permanent magnets available due to their high maximum energy product, which is the most important factor to be considered in selecting a permanent magnet. Low-carbon 1010 steel is used as a soft ferromagnetic material for conducting and focusing the magnetic flux. Although magnetically not ideal, the material was chosen for its low price and good machining properties. The structure has four air gaps with a strong magnetic field and four areas of low magnetic field where the magnetocaloric material (AMRs) circulates during the operation of the magnetic refrigerator (Figs. 1 and 2a).

After finalizing the geometry of the magnetic structure, the magnetic field generated by it was simulated. The results are presented graphically with the distribution of the magnetic flux density and with a graph of the magnetic flux density (B) as a function of the angle of the AMR rotation (Fig. 2).

2.2. Measurement of magnetic flux density

To verify the analyses or the simulation of the magnetic field generated by the presented magnetic structure, the research

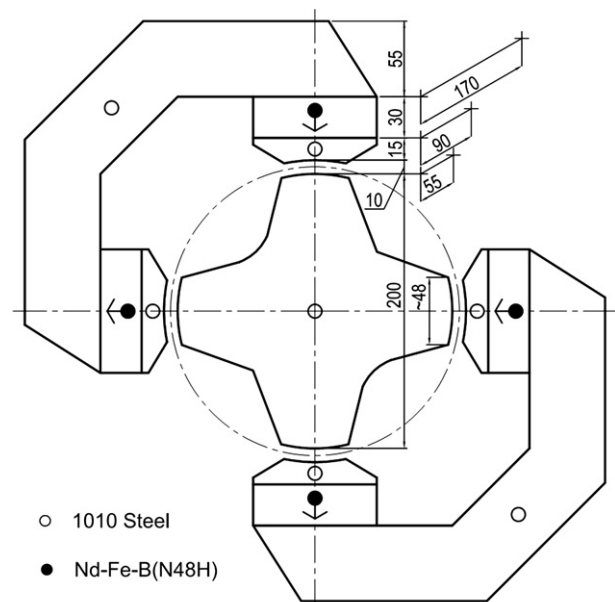


Fig. 1 – Scheme of magnetic structure with optimized dimensions (in 2D form).

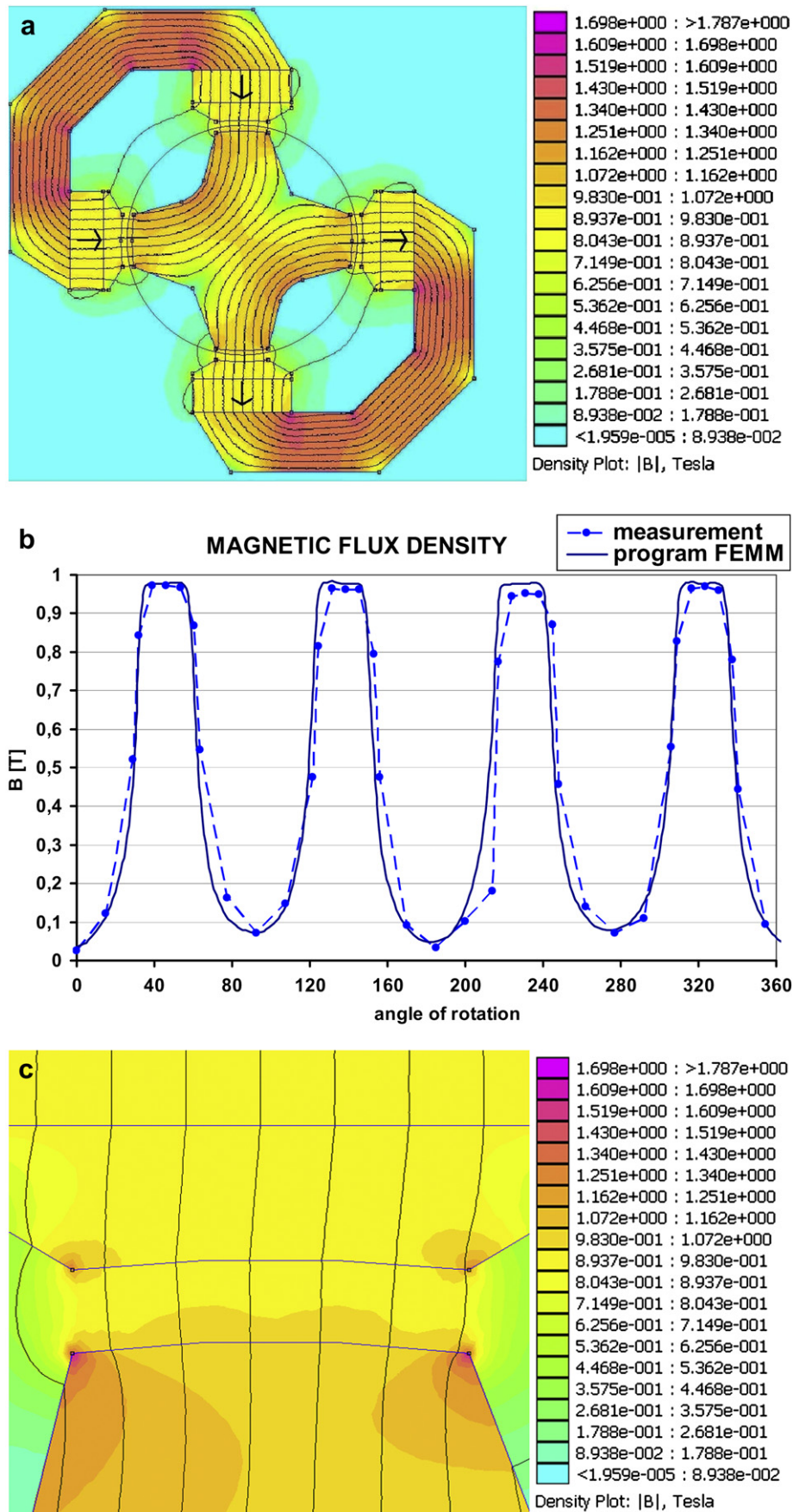


Fig. 2 – Magnetic flux density distribution (a), density for the AMRs pitch circle (b) and magnetic flux density air gap detail (c).

team built and assembled the prototype and, in order to study the magnetic field generated by the presented structure in detail, measured the magnetic flux density in the magnetic structure. The measurements were made using a three-axis magnetometer with an integrated three-axis Hall probe (SENIS transducer x-H3x-xx_E3D-2.5kHz-0.1-2T), which is the most appropriate instrument for such measurements on account of its small size and precision and was presented by Popovic et al. (2007). The magnetic flux density was measured at 40 measuring points arranged in such a way that three measurement points were in the middle of each air gap, two on the internal edge and two on the external edge of each air gap, while three measurement points were in each demagnetizing area, thus covering the entire circle where AMRs with the magnetocaloric material are rotating.

The obtained results are shown in Fig. 2b. For comparison, the graph also presents the results obtained using the FEMM program. Fig. 2b shows that in the air gaps where the magnetocaloric material is expected to magnetize, the magnetic flux density amounts to 0.98 T. It is also evident that in the areas where the magnetocaloric material is expected to demagnetize, the magnetic flux density is around 0.05 T, which means that the magnets and soft steel are at a sufficient distance from the expected demagnetization areas to provide a suitably low magnetic flux density.

To estimate the homogeneity of the magnetic flux density in the air gaps, which is very important for the efficient operation of the magnetic refrigerator, the magnetic flux density was measured at different heights (radial direction on the magnetic structure) and depths (axial direction on the magnetic structure) of the air gaps. It was concluded that the magnetic flux density varies considerably with the height in the air gap. In the middle of the gap, the magnetic flux density is almost perfectly homogenous. However, in the vicinity of the magnets and at the largest distance from them, the magnetic flux density homogeneity is much lower and varies inside of each air gap by as much as 0.2 T (Fig. 2c). At the same time, the homogeneity of the magnetic flux density is much better for different depths of the air gaps due to the 0.95 T magnetic flux density at the front and back edges of the air gaps.

A comparison of the results obtained with the FEMM program and the measured magnetic flux density values (Fig. 2b) indicates a very good agreement of results, which

confirms the suitability of the FEMM program for estimating the magnetic field generated by magnet circuit structures.

3. Operation of the magnetic refrigeration system

The operation of the rotary magnetic refrigeration (MR) system is based on AMRs rotating through air gaps and through demagnetization areas, as shown in Figs. 2 and 3.

The scheme and the picture of the magnetic refrigeration system are given in Figs. 3 and 4. The basic operating principle of the system is quite complex (regarding to Fig. 4). The fluid (distilled water) flows from the hot reservoir I (6) through the demagnetized AMRs situated in demagnetization areas at the time to the cold reservoir II (7). From there the fluid flows through the cold heat exchanger to the cold reservoir I (8) and further through the magnetized AMRs situated in the air gaps at the given moment to the hot reservoir II (5). The fluid then flows through the expansion vessel (13) and the pump (14) to the hot heat exchanger. A circuit is completed when the fluid returns (through filter – 16) to the hot reservoir I. The fluid flow is controlled by a sliding flow divider (4) which uses relative movement of the two surface areas to direct the fluid flow through regularly placed chambers with AMRs, depending on their position in the magnet structure.

Fig. 5 presents a detail of the rotating drum into which AMRs are placed and which rotates through air gaps and demagnetization areas. In the first stage of the research gadolinium (Gd) is used, a material that has established itself as a prototype magnetocaloric material. AMRs are made of Gd plates (thickness of 0.3 mm) linked into plate packs with spacers that ensure gaps for fluid flow and porosity of 0.59. This AMR geometry is used due to the thermo-hydraulic characteristics (see Šarlah, 2008) of the then available form and state of Gd. For other technical properties of the MR see Table 1.

4. Concept and design of magnetic refrigerator

The primary goal in designing our first prototype was to make the central part of MR as small and compact as

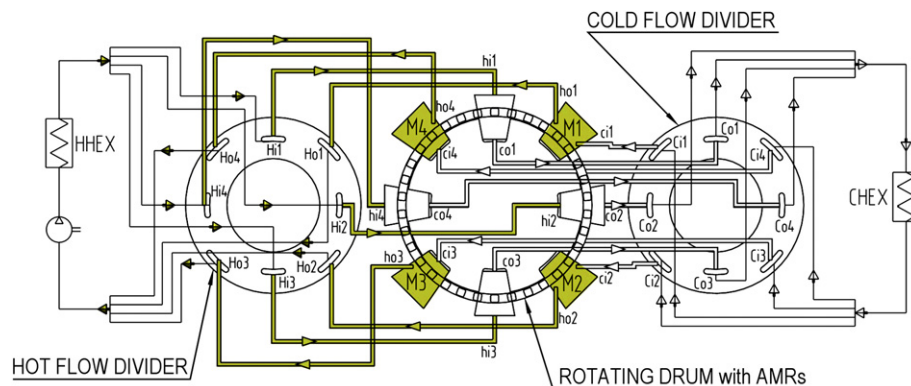


Fig. 3 – Flow diagram of the MR with the hot (left) and cold (right) flow dividers and four magnetization (M1...M4) and demagnetization areas.

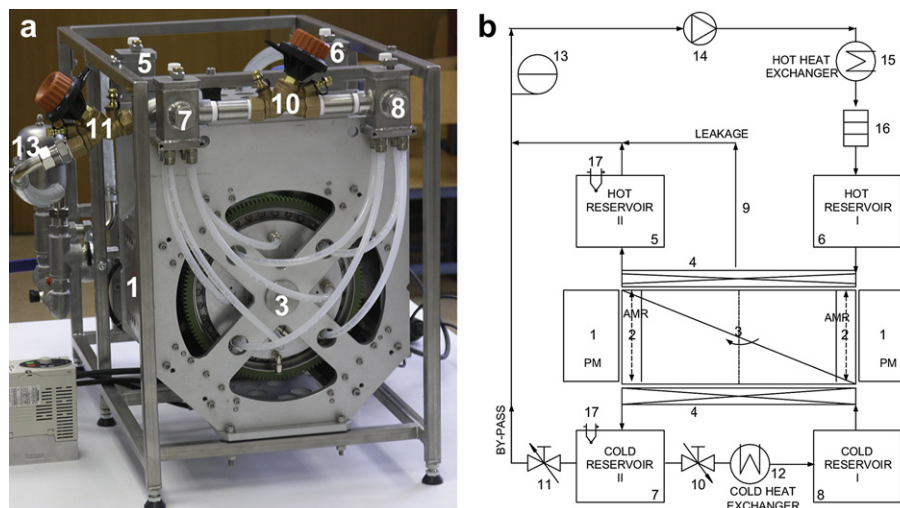


Fig. 4 – Magnetic refrigerator (MR): picture (a) and scheme (b).

possible and at the same time to ensure that all its parts could be fabricated to high-quality standards and under general toolmaking principles using standard machining methods. In short, our aim was to avoid any demanding and/or costly material conversion procedures. Nevertheless, the design of the magnetic refrigerator would have to allow for subsequent redesigning in the process of which the components could be redesigned to meet the requirements for mass production mainly through the use of conversion procedures and unconventional materials (polymers, ceramics).

The rotor containing a magnetocaloric material (AMRs) is drum-shaped and rotates around the static axis, so the AMRs rotate through the magnetization and demagnetization areas (Figs. 3 and 6b). In order to minimize the thickness of its wall and chambers, i.e. to ensure maximum cross-section of the AMRs, the drum is made from Al-alloy and the chambers for

AMRs were cut using the Wire Electrical Discharge Machining (WEDM) method.

The heat transfer fluid flows, alternately, through the drum in the axial direction (Fig. 7). In order to control the fluid flow between the channels with AMRs, a fluid flow divider is needed on each side of the drum. The dividers were designed so as to take up the minimum width of the system and minimize the fluid channel length. To achieve this, their shape resembled axial journal bearings where the contact force is controlled through a set of pressure springs. Stainless steel (rotating area – part of the frontal wall of the drum) and PTFE coating glued and processed on the base Al-alloy plate (still area) are used as a sliding pair. The rotating and still parts are sealed using standard rubber radial shaft seals (Fig. 7), whereas the drum with AMRs and the frontal plate of the drum (polyamide) are sealed using laser-cut flat seals compressed with a bolt connection.

With a view to reduce the heat transfer through the refrigerator parts, the design of the drum and the flow dividers is such that it minimizes the length of fluid channels (Fig. 7) and allows the highest possible number of parts to be fabricated from low thermal conductivity materials (polyamide, PTFE).

The main limitation during the process came from the WEDM technology of manufacturing the AMR chambers (which was the only one available during the prototype design process). This is also the main reason why the rotating drum is made from Al-alloy, which in terms of thermal conductivity is not the most favorable choice of material.

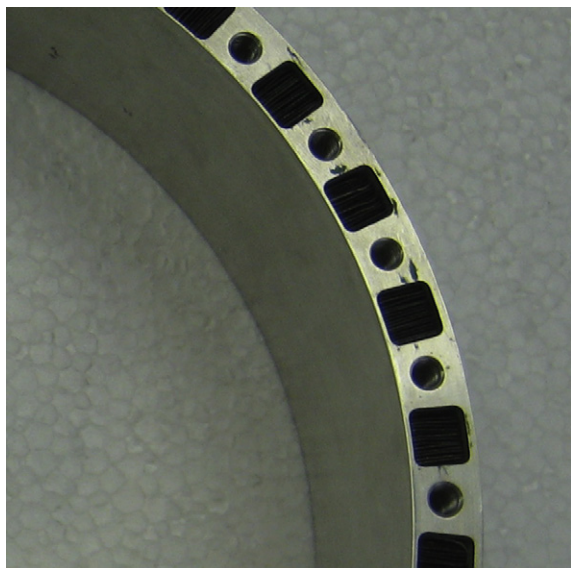


Fig. 5 – A detail of the drum with chambers for AMRs.

Table 1 – Technical properties of the MR.

Magnetic flux density	0.05 T–0.98 T
Operating frequency	0.25 Hz–4 Hz
Heat transfer fluid	Distilled water
Number of AMRs	34
Magnetocaloric material	Gd
Mass of magnetocaloric material	~0.6 kg
Dimensions of AMR	10 mm × 10 mm × 50 mm

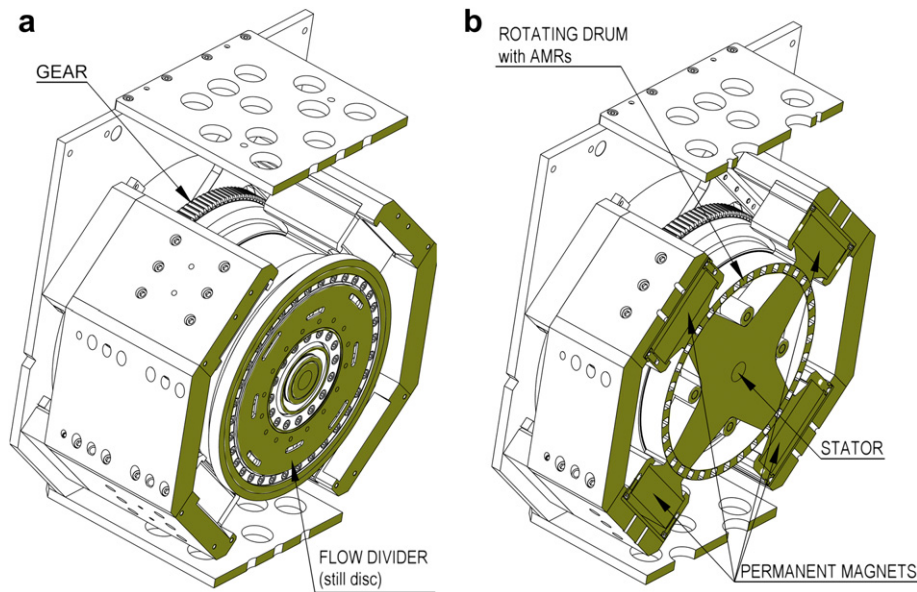


Fig. 6 – Magnetic refrigerator cross-section through the flow divider (a) and through the drum with AMRs, inner stator and magnets (b).

The drum is driven by a gear transmission system. The pinion ($z_1 = 23$) is mounted directly on the output shaft of the frequency controlled 8 pole asynchronous electromotor ($n_{EM} = 700$ rpm), and the toothed ring ($z_2 = 123$) is fitted on the ring of the drum top plate. During the testing, we came to a conclusion that the performance of the magnetic refrigerator is highest at the lowest level of the stable and permanently achievable rotational speed of the drum (~ 60 rpm) which,

considering the restrictions of the toothed ring (gear ratio and dimensions) and the frequency controlled electric motor (EM), can be achieved. For experimental purposes, we wish to have a bigger gear ratio of the gear pair which would enable us to determine optimal working point of the refrigerator.

After assembly, the performance of the construction was within expectations. However, several disadvantages of the construction became evident. Any generation of heat and excess thermal conductivity of component parts causes problems or thermal losses, which was expected. Several problems arose, however, that were connected to machining technology. The stability of the shape and dimensions and the precision of polyamide-made parts were found to be in close connection with temperature and humidity absorption. Heat generation is known to severely affect the maximum cutting precision in the process of machining. During the operation of MR, the condition deteriorated further largely due to the presence of fluid (humidity) and local heat generation (friction). The key parts of the system should therefore be made from time- and temperature-stable materials with low thermal conductivity (ceramics).

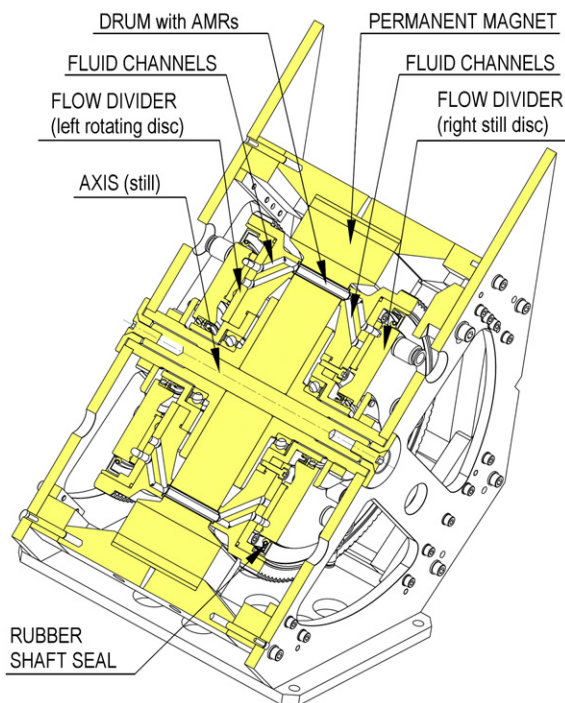


Fig. 7 – Crossection along drum axis.

Standard shaft seals are a financially attractive option that can be used in the production of prototype but this is functionally inappropriate. Owing to its size, the outer seal is much too robust, causing excess rotational resistance despite tolerance adjustments. Similar problems arose with regard to the sliding divider (large cross-section and surface – Fig. 6a). The above mentioned issues were considered in determining engine power but the friction in seals and sliding flow dividers generates a substantial amount of heat that reduces the efficiency of the refrigerator. Reduced friction (lower clamping force) leads to an increased loss of fluid (leakage), which returns to the system but still negatively affects the performance of the refrigerator. The research showed that the selected materials and machining technologies make it hard

Table 2 – Pros and cons of the developed magnetic refrigerator.

PROS	CONS
Four processes of magnetization and demagnetization in one rotation of the drum assure higher operation frequency of AMR.	Problems with sealing causes rotational resistance, heat generation (flow divider and large diameter shaft seals) and relatively big leakage.
Minimized length of the fluid channels due to the “axial flow divider” – compact design.	Relatively complex mechanical structure (costs of manufacture, long assembly and disassembly times) and large weight of magnetic structure.
Simple gear pair transmission system to drive the drum.	Usage of high thermal conductivity materials (i.e. drum – Al-alloy).

to achieve the desired accuracy of the form (flatness) of contact surfaces of the sliding divider, which would allow the divider to operate at a suitably low clamping force (and consequently lower friction) and with minimum leakage.

5. Conclusion

The experience obtained in the development of the first prototype magnetic refrigerator to date has pointed to the positive and negative aspects of the project (Table 2) and introduced several new ways and ideas. Among positive aspects we would especially like to point out a compact design of the machine which due to the axial flow divider assures minimized length of the fluid channels and the fact that AMRs in the machine can operate at higher operation frequency because of the four processes of magnetization and demagnetization in one rotation of the drum. On the other hand, the negative aspects of the project are most of all problems with sealing, since flow divider and large diameter shaft seals cause rotational resistance (heat generation in the system) and relatively big leakage. One of the main problems of the used concept also lies in large dimensions of the connecting structure (made from soft ferromagnetic steel) for conducting the magnetic flux between permanent magnets. The size of the structure itself is linked to large weight, which in the given concept cannot be significantly reduced. As the magnetic refrigerator needs a large quantity of magnetic material to operate, the total weight of the device will continue to be the main problem, especially when comparing the magnetic refrigerator with classic refrigeration devices. Another issue to be addressed is a relatively demanding and complex structure of the refrigerator which increases production costs and hinders assembly and subsequent maintenance.

However, certain deficiencies observed will be eliminated through changes and partial reconstructions of the existing prototype. Most of the results obtained and new ideas emerging in the course of the development process will be utilized in the manufacture of a new prototype magnetic refrigerator which is already in its design phase. In the new prototype, we will try to maintain all positive and improve all negative aspects presented in Table 2. First of all, we will use a new concept of a flow divider and through the usage of a thin fluid layer we hope to nullify rotational resistance (heat generation) and leakage in the system. But at the same time, this new idea could result in too much mixing of the fluid inside the divider. We will also use a different and smaller structure for generating magnetic field which will also assure easier assembly and disassembly of the system and lower manufacture costs. Finally, we are planning to use lower thermal conductivity materials in the housing of the machine to reduce heat transfer to the surroundings.

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